Two-Level Photonic Recognition and Temporally Directed Secret Recovery

A Theoretical Investigation of Quantum-Noise Cancellation, Photon-Distribution Recognition, and Non-Exponential Candidate Reduction

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Abstract

This paper proposes a two-level computational architecture in which photon-distribution statistics, quantum shot noise, digital noise cancellation, and temporally directed information flow are formulated as components of a unified secret-recognition process. The central hypothesis is that an optoelectronic circuit incorporating a logical AND-gate constraint followed by a noise-cancellation filter exhibits persistent statistical correlations that carry mutual information about an otherwise unknown binary secret $K \in \{0,1\}^n$. Rather than treating photon detection events as independent random outputs, the architecture models their collective empirical distribution across m measurement cycles and decodes them via a two-stage estimator. Level 1 accumulates binary observations to construct an upper-bound support set U_m comprising bit indices that have never registered a zero measurement. Level 2 performs bounded minimum-Hamming-distance decoding around the support indicator vector B_m, enumerating candidate bitstrings in increasing order of Hamming weight k subject to a cryptographic commitment $C = H(s \parallel K)$. We formalize the channel model as an asymmetric Z-channel with false-positive leakage, derive exact finite-sample tail bounds via Chernoff-Hoeffding and binomial concentration inequalities, establish the cumulative oracle query complexity across sequential observation stages, and prove information-theoretic lower bounds via Fano's inequality. Finally, we specify experimental protocols and statistical controls required to distinguish genuine physical information transfer from simulation artifacts.

1 Introduction

Conventional unstructured search algorithms attempt to identify an unknown binary string $K = (K_1, K_2, \ldots, K_n) \in \{0,1\}^n$ by evaluating candidate states across the full configuration space $\Omega = \{0,1\}^n$, requiring $|\Omega| = 2^n$ worst-case oracle queries. The exponential scaling of $|\Omega|$ renders brute-force enumeration intractable for cryptographic bit lengths.

The architecture investigated here reframes secret recovery as a statistical inference problem over a noisy physical channel. Instead of querying candidates blindly, we examine whether the physical dynamics of a photonic circuit coupled to a digital noise-cancellation stage induce a recognizable statistical bias prior to discrete candidate enumeration.

Proposition 1.1 (Fundamental Hypothesis)**.** *A photonic measurement channel constrained by a nonlinear logical AND operation and processed by a correlated noise-cancellation filter induces a joint distribution $P(\mathbf{X}^{(1:m)} \mid K)$ whose empirical support converges toward the true support of K, enabling sub-exponential candidate reduction under bounded decoding budgets.*

This framework decomposes the system into three coupled mathematical subsystems:

1. **Photonic Information Generation:** A quantum-optical sampling stage converting physical optical states into discrete binary sequences $\mathbf{X}(t) \in \{0,1\}^n$.
2. **Noise Cancellation and Temporal Transfer:** A signal-processing filter that estimates and subtracts environmental and detector fluctuations while potentially preserving cross-channel correlations.
3. **Two-Level Statistical Decoding:** A support-estimation stage (Level 1) paired with a Hamming-ball search verifier (Level 2).

2 The Photonic Computational Architecture

2.1 Photon Generation and Stochastic Process Representation

Let $(\Omega_{\text{phys}}, \mathcal{F}, \mathbb{P})$ be the underlying probability space governing the optical apparatus. At discrete measurement cycle $t \in \{1, 2, \ldots, m\}$, photodetectors across n spatial or temporal modes produce a binary random vector:

$$\mathbf{X}(t) = \big(X_1(t), X_2(t), \ldots, X_n(t)\big) \in \{0,1\}^n, \tag{1}$$

where $X_i(t) = 1$ denotes a threshold-exceeding photon detection event in mode i during cycle t, and $X_i(t) = 0$ denotes a null detection. The complete experimental record over m cycles forms a random matrix $\mathbf{X}^{(1:m)} \in \{0,1\}^{m \times n}$.

2.2 Quantum Noise and Photodetection Statistics

In quantum optics, a coherent state $|\alpha_i\rangle$ with mean photon number $\mu_i = |\alpha_i|^2$ incident on a single-photon detector with quantum efficiency $\eta \in (0,1]$ and dark-count rate ν_d yields photon counts governed by a Poisson distribution:

$$\mathbb{P}(N_{\text{ph},i} = c) = \frac{(\eta\mu_i + \nu_d)^c e^{-(\eta\mu_i + \nu_d)}}{c!}, \quad c \in \mathbb{N}_0. \tag{2}$$

For an ideal threshold detector ($\nu_d \to 0$) firing whenever $c \geq 1$, the marginal detection probability in mode i is:

$$p_i = \mathbb{P}(X_i(t) = 1) = 1 - \mathbb{P}(N_{\text{ph},i} = 0) = 1 - e^{-\eta\mu_i}. \tag{3}$$

In the homogeneous regime where optical power and detector efficiency are uniform across all n channels ($\mu_i = \mu$), we obtain $p_i = p = 1 - e^{-\eta\mu}$ for all $i \in \{1, \ldots, n\}$. In practical hardware, spatial crosstalk and detector dead time introduce cross-mode covariance $\text{Cov}(X_i(t), X_j(t)) \neq 0$ and temporal autocorrelation $\text{Cov}(X_i(t), X_i(t+\tau)) \neq 0$, which must be characterized empirically.

2.3 The AND-Gate Logical Constraint

The photonic architecture incorporates a nonlinear logical AND interaction between two binary optical or optoelectronic signals $A, B \in \{0,1\}$, defined by the Boolean product $Y = A \wedge B = A \cdot B$.

Under independent Bernoulli inputs $A \sim \text{Bernoulli}(q_A)$ and $B \sim \text{Bernoulli}(q_B)$, the output is $Y \sim \text{Bernoulli}(q_A q_B)$. When $q_A = q_B = 1/2$, the gate introduces a pronounced statistical asymmetry:

$$\mathbb{P}(Y = 1) = \frac{1}{4}, \qquad \mathbb{P}(Y = 0) = \frac{3}{4}. \tag{4}$$

Table 1: Truth table and probability mass distribution of the AND-gate constraint under independent Bernoulli inputs $A \sim \text{Bernoulli}(q_A)$ and $B \sim \text{Bernoulli}(q_B)$.

Input A	**Input** B	**Output** $Y = A \wedge B$	**General Probability**	**Uniform (**$q_A = q_B = 1/2$**)**
0	0	0	$(1 - q_A)(1 - q_B)$	$1/4$
0	1	0	$(1 - q_A)q_B$	$1/4$
1	0	0	$q_A(1 - q_B)$	$1/4$
1	1	1	$q_A q_B$	$1/4$

Crucially, if A represents a fixed secret bit $K_i \in \{0, 1\}$ and B represents a stochastic photonic probe $R_i(t) \sim \text{Bernoulli}(p)$, the AND gate acts as a multiplicative mask: $Y_i(t) = K_i \cdot R_i(t)$.

3 The Noise-Cancellation Hypothesis

3.1 Noise Cancellation as an Information-Bearing Transformation

Conventional signal processing views noise cancellation purely as variance reduction. However, because any physical or digital noise estimator $\widehat{N}(t)$ is computed from measurements coupled to the optical apparatus, its internal state can retain statistical dependencies on the underlying physical interaction.

3.2 Continuous Signal Decomposition and Residual Covariance

Let the raw analog signal prior to thresholding be modeled as an additive stochastic process:

$$S(t) = I(t) + N_q(t) + N_d(t), \tag{5}$$

where $I(t)$ is the secret-modulated optical signal, $N_q(t)$ is zero-mean quantum shot noise with variance σ_q^2, and $N_d(t)$ is electronic/detector noise with variance σ_d^2. A digital noise-cancellation filter constructs an estimate $\widehat{N}(t)$ of the total noise $N(t) = N_q(t) + N_d(t)$ and outputs the compensated signal:

$$S_c(t) = S(t) - \widehat{N}(t) = I(t) + \tilde{N}(t), \qquad \tilde{N}(t) \triangleq N(t) - \widehat{N}(t). \tag{6}$$

The variance of the compensated signal expands as:

$$\text{Var}(S_c) = \text{Var}(I) + \text{Var}(\tilde{N}) + 2\,\text{Cov}(I, \tilde{N}). \tag{7}$$

When $\widehat{N}(t)$ is derived from a reference channel correlated with $I(t)$, the cross-covariance term $\text{Cov}(I, \tilde{N}) = \text{Cov}(I, N) - \text{Cov}(I, \widehat{N})$ becomes non-zero, altering the effective signal-to-noise ratio (SNR) and shifting the post-threshold binary transition probabilities.

3.3 Mutual Information and Data Processing Inequality

To quantify whether the cancellation stage preserves secret-dependent structure, we evaluate the Shannon mutual information between the secret K and the compensated output S_c:

$$\mathcal{I}(K; S_c) = H(K) - H(K \mid S_c) = \sum_{k \in \{0,1\}^n} \int_{\mathcal{S}} p(k, s_c) \log_2 \frac{p(k, s_c)}{p(k)p(s_c)}\, ds_c. \tag{8}$$

By the Data Processing Inequality (DPI), for any Markov chain $K \to S \to S_c$ where S_c is a deterministic or randomized function of S alone without side-channel access to K, we must have:

$$\mathcal{I}(K; S_c) \leq \mathcal{I}(K; S). \tag{9}$$

Equality holds if and only if S_c is a sufficient statistic for K relative to S. However, if the noise-cancellation circuit incorporates an auxiliary reference channel W correlated with the environmental noise N (i.e., $K \to (S, W) \to S_c$), conditioning on W can increase accessible mutual information relative to the uncompensated single-channel measurement:

$$\mathcal{I}(K; S_c) = \mathcal{I}(K; S - \widehat{N}(W)) > \mathcal{I}(K; S) \quad \text{whenever } \mathcal{I}(K; W \mid S) > 0. \tag{10}$$

Proving $\mathcal{I}(K; S_c) > 0$ experimentally is necessary to establish statistical dependence, though not by itself sufficient to guarantee polynomial-time reconstruction.

3.4 Circuit-Tracing Signal Pathway

The complete physical-computational cascade is modeled as the directed composition of stochastic maps:

$$\text{Photon Generation} \xrightarrow{\mathcal{E}_{\text{opt}}} \text{Photonic Interaction} \xrightarrow{\mathcal{E}_{\text{AND}}} \text{AND Constraint} \xrightarrow{\mathcal{E}_{\text{NC}}} \text{Noise Cancellation} \xrightarrow{\mathcal{D}} \text{Recognitio} \tag{11}$$

Experimental validation requires measuring $\mathcal{I}(K; \cdot)$ at each intermediate node to verify that information is neither destroyed by saturation nor artificially injected by software leakage.

4 Temporal Directionality

4.1 One-Directional Causal Information Flow

The architecture enforces a strictly causal, forward-directed processing pipeline without invoking retrocausality:

$$\boxed{\text{Photon Input } \mathbf{R}(t) \longrightarrow \text{AND Gate } (K \wedge \mathbf{R}(t)) \longrightarrow \text{Noise Cancellation } S_c(t) \longrightarrow \text{Secret Recovery } \widehat{K}} \tag{12}$$

Here, temporal directionality denotes the sequential conditioning of downstream observables on upstream physical states.

4.2 Bayesian Formulation of the Secret as a Terminal Constraint

Let $K \in \{0,1\}^n$ be drawn from a prior distribution $\pi(K)$ (typically uniform, $\pi(K) = 2^{-n}$). Given m measurement vectors $\mathbf{X}^{(1:m)}$, the posterior probability of any candidate secret $\kappa \in \{0,1\}^n$ is given by Bayes' theorem:

$$\mathbb{P}\big(K = \kappa \mid \mathbf{X}^{(1:m)}\big) = \frac{\mathbb{P}\big(\mathbf{X}^{(1:m)} \mid K = \kappa\big)\, \pi(\kappa)}{\sum_{\kappa' \in \{0,1\}^n} \mathbb{P}\big(\mathbf{X}^{(1:m)} \mid K = \kappa'\big)\, \pi(\kappa')}. \tag{13}$$

In this formulation, K acts as the latent parameter indexing the likelihood function $\mathcal{L}(\kappa) = \mathbb{P}(\mathbf{X}^{(1:m)} \mid K = \kappa)$. The decoder approximates the Maximum A Posteriori (MAP) estimator $\widehat{K}_{\text{MAP}} = \arg\max_\kappa \mathbb{P}(K = \kappa \mid \mathbf{X}^{(1:m)})$ by restricting search to high-likelihood regions of $\{0,1\}^n$.

5 Two-Level Recognition Codes

5.1 Level 1: Asymmetric Z-Channel and Never-Seen-Zero Recognition

We formalize the idealized observation model as an asymmetric binary channel (Z-channel) acting independently across bit positions $i \in \{1, \dots, n\}$ and cycles $t \in \{1, \dots, m\}$:

$$X_i(t) = K_i \wedge R_i(t), \qquad R_i(t) \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p), \quad p \in (0,1). \tag{14}$$

Note carefully the asymmetric transition law of this channel:

$$\mathbb{P}(X_i(t) = 1 \mid K_i = 0) = 0, \qquad \mathbb{P}(X_i(t) = 1 \mid K_i = 1) = p, \qquad \mathbb{P}(X_i(t) = 0 \mid K_i = 1) = 1 - p. \tag{15}$$

For each bit index $i \in \{1, \dots, n\}$, define the cumulative zero-count statistic after m cycles:

$$Z_i(m) = \sum_{t=1}^{m} \mathbb{I}\big[X_i(t) = 0\big], \tag{16}$$

where $\mathbb{I}[\cdot]$ denotes the indicator function.

Definition 5.1 (Never-Seen-Zero Support Set)**.** *The Level-1 empirical support set $U_m \subseteq \{1, \dots, n\}$ is the set of indices that have never produced a zero observation across m cycles:*

$$U_m \triangleq \big\{i \in \{1, \dots, n\} : Z_i(m) = 0\big\} = \left\{i \in \{1, \dots, n\} : \prod_{t=1}^{m} X_i(t) = 1\right\}. \tag{17}$$

Lemma 5.2 (Support Containment and False-Positive Persistence)**.** *Under the observation model $X_i(t) = K_i \wedge R_i(t)$, let $\text{supp}(K) = \{i : K_i = 1\}$ denote the true support of K. Then for every $m \geq 1$:*

1. $\text{supp}(K) \subseteq U_m$ *with probability* 1 *(zero false negatives).*
2. *For any true zero index $i \notin \text{supp}(K)$, the probability of persisting in U_m as a false positive is:*

$$\mathbb{P}(i \in U_m \mid K_i = 0) = \prod_{t=1}^{m} \mathbb{P}(R_i(t) = 1) = p^m. \tag{18}$$

Remark 5.3 (Reconciliation of Bit Polarity Convention)**.** *In the idealized model $X_i(t) = K_i \wedge R_i(t)$, $K_i = 1$ forces $X_i(t) = R_i(t)$ (so $i \in U_m$ with probability p^m), whereas $K_i = 0$ forces $X_i(t) = 0$ identically (so $i \notin U_m$ with probability 1). Under the dual negated-input convention $X_i(t) = \neg K_i \wedge R_i(t)$ (or equivalently $X_i(t) = 1 \oplus (K_i \wedge \neg R_i(t))$), true zeros ($K_i = 0$) never produce a zero observation ($\mathbb{P}(i \in U_m \mid K_i = 0) = 1$), while true ones ($K_i = 1$) remain in U_m with probability p^m. Both formulations are isomorphic under bitwise complementation $K \mapsto \mathbf{1} \oplus K$; in what follows, we adopt the dual convention so that U_m forms an upper-bound mask containing all true zeros plus Q_m unresolved true ones.*

5.2 Upper-Bound Candidate Construction

Define the Level-1 binary indicator vector $B_m \in \{0,1\}^n$ component-wise by:

$$B_m(i) = \mathbb{I}[i \in U_m] = \begin{cases} 1, & i \in U_m, \\ 0, & i \notin U_m. \end{cases} \tag{19}$$

Under the dual Z-channel convention, whenever $B_m(i) = 0$, it is certain that $K_i = 1$ (since $K_i = 0 \implies i \in U_m$). Conversely, positions with $B_m(i) = 1$ partition into true ones ($K_i = 1$,

which will eventually be eliminated as m grows) and true zeros ($K_i = 0$, which persist in U_m permanently). Thus B_m dominates K bitwise ($K_i \le B_m(i)$ for all i), and the Hamming distance $d_H(B_m, K)$ equals the exact number of false-positive positions Q_m:

$$Q_m \triangleq d_H(B_m, K) = \sum_{i=1}^{n} \big(B_m(i) - K_i\big) = |U_m| - \big(n - w_H(K)\big), \tag{20}$$

where $w_H(K) = \sum_{i=1}^{n} K_i$ is the Hamming weight of K.

5.3 Expected Residual Uncertainty and Sample Complexity

When $K \sim \text{Uniform}(\{0,1\}^n)$, each bit independently satisfies $\mathbb{P}(K_i = 1) = 1/2$. The unconditional probability that index i is an unresolved true one at cycle m is:

$$\theta_m \triangleq \mathbb{P}(K_i = 1 \text{ and } i \in U_m) = \mathbb{P}(K_i = 1)\,\mathbb{P}(i \in U_m \mid K_i = 1) = \frac{p^m}{2}. \tag{21}$$

By linearity of expectation, the expected number of unresolved positions is:

$$\mathbb{E}[Q_m] = \sum_{i=1}^{n} \theta_m = \frac{n}{2} p^m. \tag{22}$$

For $p = 1/2$, $\mathbb{E}[Q_m] = n \cdot 2^{-(m+1)}$. Setting $\mathbb{E}[Q_m] \le \epsilon$ yields the logarithmic observation threshold:

$$m \ge \frac{\ln(n/(2\epsilon))}{\ln(1/p)} = \log_{1/p}\Big(\frac{n}{2\epsilon}\Big), \tag{23}$$

demonstrating that $O(\log n)$ measurement cycles suffice in expectation to collapse the residual uncertainty to $O(1)$ bits.

6 Level 2: Hamming-Weight Recognition

6.1 Hamming-Ball Enumeration Principle

Let $u_m = |U_m| = w_H(B_m)$ be the cardinality of the Level-1 support set. Because $K_i \le B_m(i)$ for all i, reconstructing K from B_m requires flipping a subset $F \subseteq U_m$ of cardinality $|F| = Q_m$ from 1 to 0. Representing F by its indicator vector $\mathbf{e}_F \in \{0,1\}^n$ supported on U_m, the candidate secret is:

$$\widehat{K}(F) = B_m \oplus \mathbf{e}_F, \qquad \text{supp}(\mathbf{e}_F) = F \subseteq U_m. \tag{24}$$

Rather than searching all 2^{u_m} subsets of U_m arbitrarily, Level 2 stratifies the search into concentric Hamming spheres $\mathcal{S}_k(B_m) = \{B_m \oplus \mathbf{e}_F : F \subseteq U_m, |F| = k\}$ for $k = 0, 1, 2, \ldots, r$. The cumulative number of candidates within Hamming radius r of B_m is the volume of the restricted Hamming ball:

$$N(u_m, r) = \sum_{k=0}^{r} \binom{u_m}{k}. \tag{25}$$

By standard entropic bounds on partial binomial sums, for $r \le u_m/2$:

$$N(u_m, r) \le 2^{u_m H_2(r/u_m)} \le \Big(\frac{e u_m}{r}\Big)^r, \tag{26}$$

where $H_2(\delta) = -\delta \log_2 \delta - (1-\delta)\log_2(1-\delta)$ is the binary entropy function. Whenever $r = O(1)$ while $u_m \approx n/2$, $N(u_m, r) = O(n^r)$ grows polynomially in n, yielding an exponential reduction relative to $2^{u_m} \approx 2^{n/2}$.

6.2 Monotonicity of Likelihood Ordering

Proposition 6.1 (Optimality of Increasing Hamming-Weight Search)**.** *Assume* $K \sim \text{Uniform}(\{0,1\}^n)$ *and* $p \in (0,1)$*. Among all candidates* $\kappa \in \{0,1\}^n$ *consistent with the support constraint (*$\kappa_i = 1$ *for all* $i \notin U_m$*), the posterior probability* $\mathbb{P}(K = \kappa \mid \mathbf{X}^{(1:m)})$ *is a strictly decreasing function of the Hamming distance* $k = d_H(B_m, \kappa)$*.*

Proof. For any valid candidate κ with $\kappa_i \le B_m(i)$, let $k = d_H(B_m, \kappa) = |\{i \in U_m : \kappa_i = 0\}|$ be the number of true zeros in U_m, so that $u_m - k$ positions in U_m have $\kappa_i = 1$. Since all $n - u_m$ positions outside U_m must have $\kappa_i = 1$, the likelihood of observing U_m given $K = \kappa$ is:

$$\mathbb{P}(U_m \mid K = \kappa) = (1)^k \cdot (p^m)^{u_m - k} \cdot (1 - p^m)^{n - u_m} = p^{m(u_m - k)}(1 - p^m)^{n - u_m}. \tag{27}$$

Taking the ratio of posterior probabilities for two consistent candidates $\kappa^{(1)}, \kappa^{(2)}$ at distances $k_1 < k_2$ under a uniform prior $\pi(\kappa) = 2^{-n}$ yields:

$$\frac{\mathbb{P}(K = \kappa^{(1)} \mid U_m)}{\mathbb{P}(K = \kappa^{(2)} \mid U_m)} = \frac{p^{m(u_m - k_1)}}{p^{m(u_m - k_2)}} = p^{-m(k_2 - k_1)} > 1, \tag{28}$$

since $0 < p < 1$ and $k_2 - k_1 \ge 1$. Hence enumerating $k = 0, 1, 2, \ldots$ tests candidates in exact decreasing order of posterior probability. □

6.3 Budget-Bounded Enumeration

To prevent exponential blowup when Q_m is large (e.g., at small m), the decoder enforces a maximum oracle budget B_{max} per decoding pass. Layer k is enumerated if and only if:

$$\sum_{j=0}^{k} \binom{u_m}{j} \le B_{\text{max}}. \tag{29}$$

Let $r^*(u_m, B_{\text{max}}) = \max\left\{r \in \{0, \ldots, u_m\} : \sum_{j=0}^{r} \binom{u_m}{j} \le B_{\text{max}}\right\}$ denote the maximum feasible search radius. Decoding at stage m succeeds if and only if $Q_m \le r^*(u_m, B_{\text{max}})$.

6.4 Cryptographic Commitment Verification

Candidate verification employs a salted cryptographic hash commitment $C = H(s \parallel K)$, where $H : \{0,1\}^* \to \{0,1\}^\lambda$ is modeled as a random oracle with digest length λ bits (e.g., $\lambda = 256$) and $s \in \{0,1\}^\ell$ is a public random salt. A candidate $\widehat{K}$ is accepted iff $H(s \parallel \widehat{K}) = C$. By the collision and second-preimage resistance of H, the false-acceptance probability across T_{total} distinct candidate evaluations is bounded by:

$$\mathbb{P}(\text{False Acceptance}) \le 1 - \left(1 - 2^{-\lambda}\right)^{T_{\text{total}}} \le \frac{T_{\text{total}}}{2^\lambda}, \tag{30}$$

which is negligible ($< 2^{-192}$) for any computationally feasible budget $T_{\text{total}} \le 2^{64}$ when $\lambda = 256$.

7 Exact Binomial Model and Tail Concentration Bounds

7.1 Conditional and Unconditional Distributions of Q_m

Conditioned on a fixed secret K of Hamming weight $w = w_H(K)$, each of the w true-one positions independently remains in U_m with probability p^m, while all $n - w$ true-zero positions

remain in U_m deterministically. Thus:

$$Q_m \mid (w_H(K) = w) \sim \text{Binomial}(w, p^m), \qquad u_m \mid (w_H(K) = w) = (n - w) + Q_m. \tag{31}$$

When $K \sim \text{Uniform}(\{0,1\}^n)$, $w_H(K) \sim \text{Binomial}(n, 1/2)$. By the law of total probability (mixture of binomials), the unconditional distribution of Q_m is exactly binomial:

$$\mathbb{P}(Q_m = k) = \sum_{w=k}^{n} \binom{n}{w} 2^{-n} \binom{w}{k} (p^m)^k (1-p^m)^{w-k} = \binom{n}{k} \left(\frac{p^m}{2}\right)^k \left(1 - \frac{p^m}{2}\right)^{n-k}, \tag{32}$$

so $Q_m \sim \text{Binomial}(n, \theta_m)$ with $\theta_m = p^m/2$. Its first two central moments and zero-error probability are:

$$\mathbb{E}[Q_m] = \frac{np^m}{2}, \qquad \text{Var}(Q_m) = \frac{np^m}{2}\left(1 - \frac{p^m}{2}\right), \qquad \mathbb{P}(Q_m = 0) = \left(1 - \frac{p^m}{2}\right)^n \approx \exp\left(-\frac{np^m}{2}\right). \tag{33}$$

7.2 Chernoff-Hoeffding Tail Bounds on Failure Probability

Theorem 7.1 (Finite-Sample Decoding Failure Bound)**.** *Suppose at observation cycle m, the budget B_{max} permits searching up to radius $r > \mu_m$, where $\mu_m = \mathbb{E}[Q_m] = np^m/2$. Writing $r = (1+\delta)\mu_m$ for $\delta > 0$, the probability that the true secret lies outside the searched Hamming ball is bounded by the multiplicative Chernoff inequality:*

$$\mathbb{P}(Q_m > r) \le \left(\frac{e^\delta}{(1+\delta)^{1+\delta}}\right)^{\mu_m} \le \exp\left(-\frac{\delta^2 \mu_m}{2+\delta}\right). \tag{34}$$

In particular, choosing $m \ge \left\lceil \frac{\ln(n/(2\mu_0))}{\ln(1/p)} \right\rceil$ ensures $\mu_m \le \mu_0$, making $\mathbb{P}(Q_m > r)$ independent of n for any fixed target radius $r > \mu_0$.

8 Computational Complexity and Information-Theoretic Limits

8.1 Sequential Cumulative Query Complexity

When the decoder runs incrementally after each cycle $m = 1, 2, \ldots, M$ until a commitment match occurs at stopping time $\tau = \min\{m \ge 1 : Q_m \le r^*(u_m, B_{\text{max}})\}$, the total number of hash evaluations is:

$$T_{\text{total}} = \sum_{m=1}^{\tau} T_m = \sum_{m=1}^{\tau} \sum_{k=0}^{r^*(u_m, B_{\text{max}})} \binom{u_m}{k} \le \tau \cdot B_{\text{max}}. \tag{35}$$

By the tail bound in Theorem 7.1, $\mathbb{E}[\tau] = O(\log n)$ whenever $B_{\text{max}} = \Omega(n^{r_0})$ for any fixed constant $r_0 \ge 1$. Consequently, the expected cumulative oracle complexity under the idealized i.i.d. Z-channel model scales as:

$$\mathbb{E}[T_{\text{total}}] = O\big(n^{r_0} \log n\big), \tag{36}$$

reducing the classical $O(2^n)$ brute-force search to polynomial time in n, provided physical sample acquisition remains polynomial.

8.2 Information-Theoretic Lower Bound via Fano's Inequality

Theorem 8.1 (Minimum Observation Requirement)**.** *Any decoder (including the two-level Hamming-weight decoder) that recovers $K \sim \text{Uniform}(\{0,1\}^n)$ with error probability $P_e = \mathbb{P}(\widehat{K} \neq K)$ from m independent channel uses must satisfy:*

$$m \geq \frac{n\big(1 - P_e\big) - 1}{C_{ch}(p)}, \tag{37}$$

where $C_{ch}(p)$ is the Shannon capacity per channel use. For the Z-channel with crossover probability p, the capacity is:

$$C_{ch}(p) = \log_2\Big(1 + (1-p)p^{\frac{p}{1-p}}\Big) \quad \textit{bits per channel use}. \tag{38}$$

Proof. By Fano's inequality, $H(K \mid \mathbf{X}^{(1:m)}) \leq H_2(P_e) + P_e \log_2(2^n - 1) \leq 1 + nP_e$. Since $H(K) = n$, the mutual information satisfies $\mathcal{I}(K; \mathbf{X}^{(1:m)}) = H(K) - H(K \mid \mathbf{X}^{(1:m)}) \geq n(1 - P_e) - 1$. By the memoryless channel coding theorem across n parallel modes and m cycles, $\mathcal{I}(K; \mathbf{X}^{(1:m)}) \leq mnC_{\text{ch}}(p)$. Dividing both sides by $nC_{\text{ch}}(p)$ yields the stated bound. □

Notice the distinction between *sample complexity* $m = \Omega(1/C_{\text{ch}}(p))$ (total bit observations $mn = \Omega(n)$) and *cryptographic query complexity* $T_{\text{total}} = O(n^{r_0} \log n)$. The physical channel must still transfer at least $\Omega(n)$ bits of mutual information to uniquely specify an n-bit secret; the architecture's advantage lies in replacing 2^n cryptographic hash evaluations with $O(n)$ physical measurements and polynomial post-processing.

8.3 Total Physical Resource Accounting

A rigorous complexity comparison must account for the complete resource tuple:

$$\mathcal{C}_{\text{total}}(n) = \mathcal{C}_{\text{photon}}(m, n) + \mathcal{C}_{\text{NC}}(m, n) + \mathcal{C}_{\text{decode}}(m, n, B_{\text{max}}) + \mathcal{C}_{\text{hash}}(T_{\text{total}}). \tag{39}$$

If the optical apparatus suffers exponential attenuation $p(n) = p_0 e^{-\alpha n}$ with circuit depth n, then $m = \Omega(e^{\alpha n} \log n)$, restoring exponential physical runtime despite polynomial hash queries. Thus, constant or inverse-polynomial transmission $p(n) = \Omega(n^{-\beta})$ is a mandatory physical requirement for scalability.

9 Physical Implementation and Experimental Validation

9.1 Experimental Architecture and Signal Tap Points

A hardware realization requires: (1) a pulsed single-photon or attenuated laser source; (2) an n-mode optical interferometer or spatial light modulator; (3) an electro-optic or nonlinear optical AND-gate array; (4) single-photon avalanche diodes (SPADs) or superconducting nanowire detectors (SNSPDs); (5) an FPGA-based digital noise-cancellation engine recording pre-filter $S_{\text{in}}(t)$ and post-filter $S_{\text{out}}(t)$ waveforms; and (6) an isolated host processor executing the Level-1/Level-2 decoder and SHA-256 commitment verifier.

9.2 Validation Suite and Controls

To isolate genuine physical information transfer from software artifacts, five falsifiable experiments must be executed:

Table 2: Experimental validation matrix and statistical test criteria.

Experiment	Configuration	Quantitative Observable & Null Hypothesis
A: Baseline	Noise cancellation disabled; fixed known test patterns.	Estimate empirical transition matrix $\widehat{P}(X_i \mid K_i)$ and spatial/temporal correlation matrix $\widehat{\Sigma}_{ij}$.
B: Cancellation	Noise cancellation enabled; identical optical input.	Test $\mathcal{I}(K; S_{\text{out}}) - \mathcal{I}(K; S_{\text{in}}) > 0$ and residual covariance $\text{Cov}(R(t), K) \neq 0$.
C: AND-Gate	Compare AND gate vs. linear pass-through control.	Verify asymmetric suppression $\mathbb{P}(Y = 1) \approx p/2$ vs. symmetric baseline via χ^2 goodness-of-fit.
D: Blind Recovery	Secret K loaded into isolated optical modulator; decoder receives only $S_{\text{out}}(t)$ and C.	Measure empirical stopping cycle $\widehat{\tau}$, hash count T_{total}, and recovery rate across 10^3 trials.
E: Scaling	Bit lengths $n \in \{16, 32, 64, 128, 256, 512\}$.	Regress $\log T_{\text{total}}(n)$ and $\log \tau(n)$ against $\log n$ to test polynomial vs. exponential scaling.

9.3 Mandatory Isolation Controls

Strict physical and cryptographic separation is enforced:

- **Zero Secret Leakage:** The decoder process executes in a sandboxed memory space without access to K, the optical modulator control bus, or the PRNG seed generating K.
- **Correlation-Aware Decoding:** If empirical cross-channel correlations $\widehat{\Sigma}_{ij} \neq 0$ violate the i.i.d. binomial assumption, the Level-2 Hamming distance metric must be replaced by the Mahalanobis or log-likelihood score $\ell(\kappa) = -\ln \mathbb{P}(\mathbf{X}^{(1:m)} \mid K = \kappa)$.
- **Matched Classical Controls:** Hardware results must be benchmarked against a software simulation driven by matched marginal probabilities $\widehat{p}_i$ and matched covariance $\widehat{\Sigma}_{ij}$ to verify whether quantum or optical effects provide any advantage beyond classical stochastic sampling.

10 Limitations and Future Work

Several theoretical and practical limitations govern the scope of this architecture:

1. **Idealized Z-Channel Assumption:** In real photodetectors, dark counts ($\nu_d > 0$) introduce two-sided errors where a true zero can sporadically register a click ($\mathbb{P}(X_i(t) = 1 \mid K_i = 0) = p_{\text{dark}} > 0$). Under two-sided noise, $Z_i(m) = 0$ eventually eliminates all bits as $m \to \infty$, requiring replacement of the hard never-seen-zero rule with a soft empirical frequency threshold $Z_i(m)/m < \gamma^*$.
2. **Simulation vs. Hardware:** Software prototypes that compute $X_i(t) = K_i \wedge R_i(t)$ in memory presuppose oracle access to K during sample generation. Demonstrating physical computational utility requires an experimental apparatus where K is encoded purely as

an unknown physical boundary condition (e.g., an unknown phase mask or optical attenuator array).

3. **Asymptotic Scaling:** Bounded search budgets (B_{max}) guarantee polynomial decoder runtime per stage but do not prevent decoding failure if physical SNR degrades exponentially with n.

11 Conclusion

This paper has established a rigorous mathematical formulation for two-level photonic recognition and temporally directed secret recovery. By modeling the photonic AND-gate interaction as an asymmetric Z-channel, we proved that the Level-1 never-seen-zero support set U_m traps all true zeros deterministically while eliminating false positives exponentially fast at rate p^m. Conditioning Level-2 Hamming-ball enumeration on U_m provably orders candidates by exact posterior likelihood, reducing cryptographic commitment evaluations from $O(2^n)$ to $O(n^{r_0} \log n)$ whenever $O(\log n)$ physical measurement cycles achieve constant residual error $\mathbb{E}[Q_m] = O(1)$. Via Fano's inequality and Chernoff concentration bounds, we delimited the exact information-theoretic and physical conditions under which this speedup holds. Future experimental work following the five-stage validation protocol will determine whether real photonic noise-cancellation hardware preserves sufficient mutual information to realize these theoretical bounds in physical systems.

A Core Mathematical Definitions and Summary Table

Table 3: Summary of formal notation, domains, and mathematical definitions.

Symbol	Domain	Formal Definition
K	$\{0,1\}^n$	Unknown n-bit binary secret vector $(K_1, \ldots, K_n)$.
$\mathbf{X}(t)$	$\{0,1\}^n$	Binary photodetection vector observed at measurement cycle $t \in \{1, \ldots, m\}$.
p	$(0,1)$	Homogeneous single-cycle detection probability $1 - e^{-\eta\mu}$.
$Z_i(m)$	$\{0, \ldots, m\}$	Cumulative count of zero observations at bit index i, $\sum_{t=1}^{m} \mathbb{I}[X_i(t) = 0]$.
U_m	$\mathcal{P}(\{1, \ldots, n\})$	Level-1 never-seen-zero support set $\{i : Z_i(m) = 0\}$ of cardinality $u_m = \lvert U_m \rvert$.
B_m	$\{0,1\}^n$	Level-1 upper-bound candidate vector with components $B_m(i) = \mathbb{I}[i \in U_m]$.
Q_m	$\{0, \ldots, n\}$	Number of unresolved false-positive positions $d_H(B_m, K) \sim \text{Binomial}(n, p^m/2)$.
$N(u_m, r)$	$\mathbb{N}$	Volume of restricted Hamming ball $\sum_{k=0}^{r} \binom{u_m}{k}$ up to radius r.
B_{max}	$\mathbb{N}$	Maximum candidate hash evaluations permitted per decoding stage.
$C_{\text{ch}}(p)$	$[0,1]$	Shannon capacity of the Z-channel, $\log_2(1 + (1-p)p^{p/(1-p)})$ bits/use.
C	$\{0,1\}^\lambda$	Salted cryptographic commitment $H(s \parallel K)$ verified independently of the optics.

B Core Equations Reference

$$\text{Quantum Detection Law:} \quad p_i = 1 - e^{-\eta\mu_i} \tag{40}$$

$$\text{Compensated Signal Variance:} \quad \text{Var}(S_c) = \text{Var}(I) + \text{Var}(\tilde{N}) + 2\,\text{Cov}(I, \tilde{N}) \tag{41}$$

$$\text{Z-Channel Observation Model:} \quad X_i(t) = K_i \wedge R_i(t), \quad R_i(t) \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p) \tag{42}$$

$$\text{Expected Residual Uncertainty:} \quad \mathbb{E}[Q_m] = \frac{np^m}{2}, \quad \text{Var}(Q_m) = \frac{np^m}{2}\left(1 - \frac{p^m}{2}\right) \tag{43}$$

$$\text{Chernoff Failure Bound:} \quad \mathbb{P}\big(Q_m > (1+\delta)\mu_m\big) \le \left(\frac{e^\delta}{(1+\delta)^{1+\delta}}\right)^{\mu_m} \tag{44}$$

$$\text{Fano Sample Lower Bound:} \quad m \ge \frac{n(1-P_e) - 1}{C_{\text{ch}}(p)} \tag{45}$$

$$\text{Cumulative Oracle Complexity:} \quad \mathbb{E}[T_{\text{total}}] = \mathbb{E}\left[\sum_{m=1}^{\tau} \sum_{k=0}^{r^*(u_m, B_{\text{max}})} \binom{u_m}{k}\right] = O(n^{r_0} \log n) \tag{46}$$